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# The Layers of Plant Responses to Insect Herbivores

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## Abstract

Plants collectively produce hundreds of thousands of specialized metabolites that are not required for growth or development. Each species has a qualitatively unique profile, with variation among individuals, growth stages, and tissues. By the 1950s, entomologists began to recognize the supreme importance of these metabolites in shaping insect herbivore communities. Plant defense theories arose to address observed patterns of variation, but provided few testable hypotheses because they did not distinguish clearly among proximate and ultimate causes. Molecular plant-insect interaction research has since revealed the sophistication of plant metabolic, developmental, and signaling networks. This understanding at the molecular level, rather than theoretical predictions, has driven the development of new hypotheses and tools and pushed the field forward. We reflect on the utility of the functional perspective provided by the optimal defense theory, and propose a conceptual model of plant defense as a series of layers each at a different level of analysis, illustrated by advances in the molecular ecology of plant-insect interactions.

**Molecular chemical ecology:** the field investigating the evolution of ecological interactions, and the functional traits and molecular and chemical mechanisms mediating these interactions

**Levels of analysis:** explanatory levels of proximate and ultimate causation

## INTRODUCTION

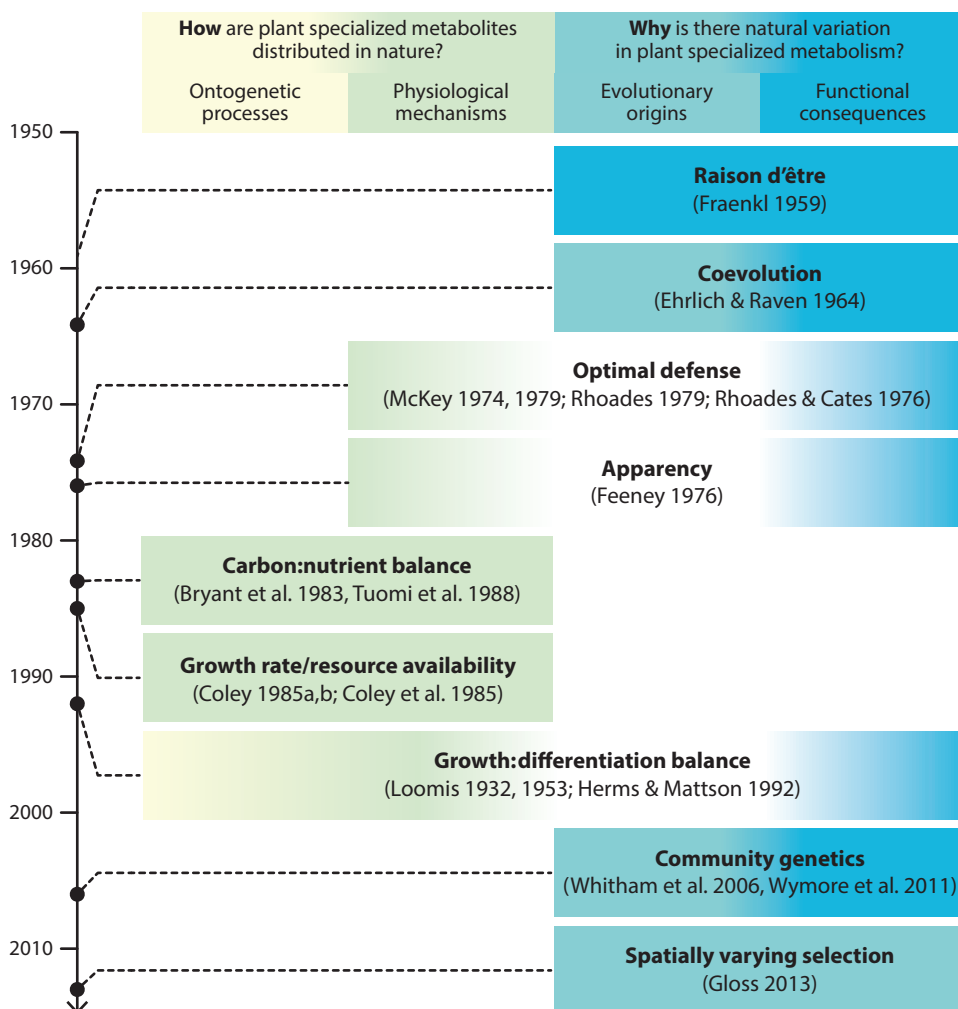
Most plant species are sessile and, during a large portion of their life histories, literally rooted to the ground. They must either adapt themselves to their environment or adapt their environment to their needs. It is therefore not surprising that plants are among nature's most accomplished chemists, producing a cornucopia of secondary, or specialized, metabolites with which they manipulate the physiology and behavior of insects and other organisms. In contrast to primary, or general, metabolites commonly required for cell growth (77), which include most carbohydrates, amino acids, and fatty acids, specialized metabolites such as alkaloids are by definition unevenly distributed, both across phylogenetic groups and among individuals within a species (e.g., 9, 32, 40, 44, 48, 63, 75, 87, 113). Of course animals and microbes also produce specialized metabolites, e.g., defensive glandular exudates or antibiotics; and in the animal kingdom there are many examples of co-opting, or sequestering specialized metabolites or their precursors, usually from food and thus often from plants (reviewed in 9, 53). This complex chemical web of life is the subject of molecular chemical ecology, which originated in studies revealing that phytochemicals can determine the occurrence and outcome of plant-herbivore interactions (reviewed in 40, 51).

Many early chemical ecologists were zoologists who were interested in the distribution and evolution of specialized plant metabolites because of the apparent consequences for distribution and evolution of insect herbivores (32, 40) (**Figure 1**, 1950s–1960s). The theories that emerged (**Figure 1**, 1970s–1990s) posited that plants encounter trade-offs between the costs of producing these metabolites, in the face of limited resources and coevolving herbivores, and their benefits via reduced herbivore pressure on plant fitness (9, 123). However, the physiological mechanism through which a trait occurs does not reveal its function, just as function does not dictate a mechanism—although each may hint at a reasonable range of possibilities for the other. The hypotheses derived from these theories were thus frequently neither testable nor mutually exclusive (8, 9, 123).

In 1988, Sherman (118, p. 616) addressed other, similar instances of “talk[ing] past each other” in biology, building on earlier work by Tinbergen (130) and Mayr (84) to point out that there are four levels at which biological phenomena can be understood: (a) Ontogenetic mechanisms and (b) physiological or cognitive processes determine how a phenomenon happens; while (c) evolutionary trajectories and (d) ecological pressures determine why it happens that way, or at all (**Figure 1**). Biological phenomena can, and ideally should, be understood at all four levels, but hypotheses may only compete and be tested within a level (118). For example, “increased production of specialized compound A entails a reduction in growth” and “the function of compound A is to defend plants attacked by insect herbivores” are not competing hypotheses. One, both, or neither may be true, and different types of data must be generated to test each: comparing growth of plants differing only in their production of compound A in the first case; and comparing fitness of plants under herbivore attack differing only in their production of compound A in the second case.

Classical plant defense theories and their insights and shortcomings have been thoroughly reviewed elsewhere (e.g., 8, 9, 20, 39, 101, 123). Here, we focus on how the utility of these theories has been limited by their failure to cleanly separate biological levels of analysis; this is both a universal problem in biology (84, 118) and a topic that has rarely been handled (but see 8). From this point of view, we critically discuss the optimal defense framework and the idea of growth-defense trade-offs, which are perhaps the most useful of the body of classical plant defense theories (8, 9, 123). We then provide a synthetic overview of the advances in molecular physiology, which, in the absence of clear guidance from theory, have driven the field forward (8, 9). Finally, we offer a perspective on maintaining clear divisions of hypotheses by level of analysis as key





**Figure 1**

Several hypotheses from different theoretical frameworks have been put forth over six decades to explain how and why observed variation in specialized plant metabolites exists. None of these frameworks attempts to cleanly integrate the levels of analysis at which biological hypotheses can be formulated (8, 118). Because the frameworks address different levels of how or why (and some blend multiple levels), they generally do not provide alternative hypotheses but rather different, to an extent complementary, points of view.

discoveries in molecular physiology are extrapolated to the level of ecological interactions, in an attempt to avoid similar problems of unfalsifiability and talking past each other in the future.

### Plant Defense Theory: What Level Are We On?

**Figure 1** portrays key concepts and frameworks in classical plant defense theory as well as selected newer frameworks in terms of how they address, or mix, levels of analysis. In 1959, Fraenkel (40) argued that plant secondary metabolites evolved primarily under ecological pressure to thwart herbivores, while herbivores in turn evolved to avoid, detoxify, or sequester these metabolites or

**Defense trait:** a trait that increases a plant's Darwinian fitness under stress such as herbivory

**OD:** optimal defense

**CNB:** carbon:nutrient balance

**GR/RA:** growth rate/resource availability

**GDB:** growth:differentiation balance

**Indirect defense:** plant defense trait that indirectly affects herbivores by attracting their enemies through provision of resources or information

even use them to identify suitable host plants. Five years later, Ehrlich & Raven (32) proposed that coevolution between plant defense and herbivore counteradaptations could explain the diversity of plants and their herbivores: Plants evolve new defense traits in response to herbivore pressure, the resulting reduction in pressure permits a radiation in plant diversity, and that radiation creates new host plants to which herbivores then adapt (adapt-and-radiate coevolution; 123, 129). Fox (39) later proposed that ephemeral plants interacting with a few, primarily specialized herbivores may undergo stepwise coevolution in a true evolutionary arms race, while more persistent plant species may undergo diffuse coevolution, developing a complex suite of defense traits to hedge against fitness loss to a large, complex, and variable herbivore community. Usually plant defense traits are taken to refer to specialized plant metabolites, but a robust definition of defense incorporates other fitness-enhancing responses (discussed below). Coevolutionary hypotheses are difficult to test because the phylogenetic distribution of traits cannot be predicted from their ecological functions, or vice versa. However, Fraenkel's (40) and Erlich & Raven's theses (32) stimulated entomologists to ask why, and how, plant defenses are distributed as they are in nature, and how this distribution has influenced the evolution, behavior, and physiology of insect herbivores.

Perhaps the most successful of these theories has been the optimal defense (OD) theory (85, 86, 102, 103), which has provided testable functional hypotheses. By quantifying plant defense in the currency of Darwinian fitness, OD provides a link to evolutionary theory (9, 25, 30). This is not true of carbon:nutrient balance (CNB) theory (15, 133), growth rate/resource availability (GR/RA) theory (21–23), or growth:differentiation balance (GDB) theory (58, 76, 77), none of which addresses functional questions (97, 123). The two main hypotheses of OD are (a) defenses evolve and are allocated to maximize individual inclusive fitness, and (b) defenses have a fitness cost because they divert resources from other processes, mainly growth and reproduction (102, 123). The first hypothesis is rather the assumption that most traits are adaptive, positing that any pattern of defense is possible as long as it is adaptive (85, 86, 102, 123). The second hypothesis is falsifiable (if imprecisely worded; see 34) and produced several subhypotheses. (a) The allocation hypothesis states that within a plant, tissues are defended in direct proportion to their fitness value and risk of predation, and in inverse proportion to the cost of allocation. (b) The plasticity hypothesis states that defenses decrease in the absence of enemies and increase in their presence. (c) The stress hypothesis states that individuals are less defended when under additional stresses because they cannot afford to allocate as many resources to defense against herbivores (36, 102, 103, 123).

The hypotheses of OD theory are compatible with apparency theory (36), and the concept of apparency is implicit in the allocation hypothesis (risk of predation) and the plasticity hypothesis (presence of enemies) (103, 123). Apparency theory (36) predicts that defenses evolve in direct proportion to predation risk and inverse proportion to fitness cost. It has largely been criticized as impractical: Although Feeney (36) defined apparency as the quality of being predictable in time and space, to quantify the apparency of a host plant or tissue, one must understand the prediction mechanism used by its herbivores, and thus apparency may be a quality of an interaction and not of a plant per se (39, 123). However, apparency theory has been remarkably useful to our understanding of the widespread phenomenon of indirect defenses, which function by increasing the apparency of herbivores to their enemies (68). The apparency and OD theories have together provided a handful of testable hypotheses posed at a functional level of analysis. Thus, among the classical plant defense theories, these likely have the most empirical support (reviewed in 89, 123). Nevertheless, the trade-off concept invoked by both OD and apparency theories may encourage researchers to confuse physiological mechanism with functional roles.

As an illustration, evidence for the stress hypothesis of OD is mixed: Plants subjected to non-herbivory stresses such as drought are predicted to be less well defended, but often are not, and

sometimes appear to be better defended (52). Admittedly, this literature is obfuscated by the challenge of stressing plants without stressing herbivores, as well as imprecise definitions of stress (73). Nevertheless, the mixed evidence is consistent with observations indicating that plant defense traits may have multiple functions, and that the expression of these traits, as well as their fitness consequences, depends on multiple ecological factors in addition to herbivore pressure (28, 36, 39). Patterns of plant defense are likely better understood as positions on a changing fitness landscape, rather than as solutions to an optimization problem (87, 98).

Similarly, observed plant phenotypes may not be best explained mechanistically by a trade-off between defense traits and growth and development. Plants excel at coordinating their metabolism interactively with their biotic and abiotic environments, both responding to their environment and changing it, to avoid situations in which resource availability would constrain their options (see, e.g., 28, 48, 74, 87, 98, 111, 145). The extent to which any defense trait limits growth and development depends on the interaction of multiple physiologically and ontogenetically regulated processes, each of which individually contributes to, but does not solely determine, the observed growth and defense phenotype. Each of these mechanisms involves some clearly defined trade-off at the level of, e.g., cell expansion or proliferation versus differentiation, allocation of nutrients, substrate channeling, or protein stability; but they may interact to buffer potential trade-offs at the ecological or functional level (74, 76, 77). In the end, the fitness costs and benefits of any trait, and whether it supports or competes with other traits' contribution to plant fitness, depend on current (ecological) and historical (evolutionary) pressures (**Figure 2**) (6, 88). Despite other failings, GDB theory recognizes that physiological trade-offs at a cellular level do not equate to trade-offs at an ecological level (74): It predicts, for example, that high levels of nutrients and light will result in high rates of both growth and defense (123).

Functional population genetics approaches have now been added to the defense theory repertoire (**Figure 1**, 2000 onward). Spatially varying selection (SVS) theory (44) draws a direct connection between the environment-dependent Darwinian fitness consequences of genetically controlled defense traits and traits' predicted frequency in a population, but it does not consider ecological functions or physiological mechanisms of these traits. Similarly, community genetics theory (145) proposes equations derived from population genetics that can be used to scale up from the function of a genetic trait in an individual to its effects in ecological communities. But although it considers ecological mechanisms cleanly divided into evolutionary and functional levels, this approach neglects physiological mechanisms of traits. Thus, neither approach provides a full biological picture, but each offers insights into how ecological contexts might be explicitly integrated into updated plant defense theory.

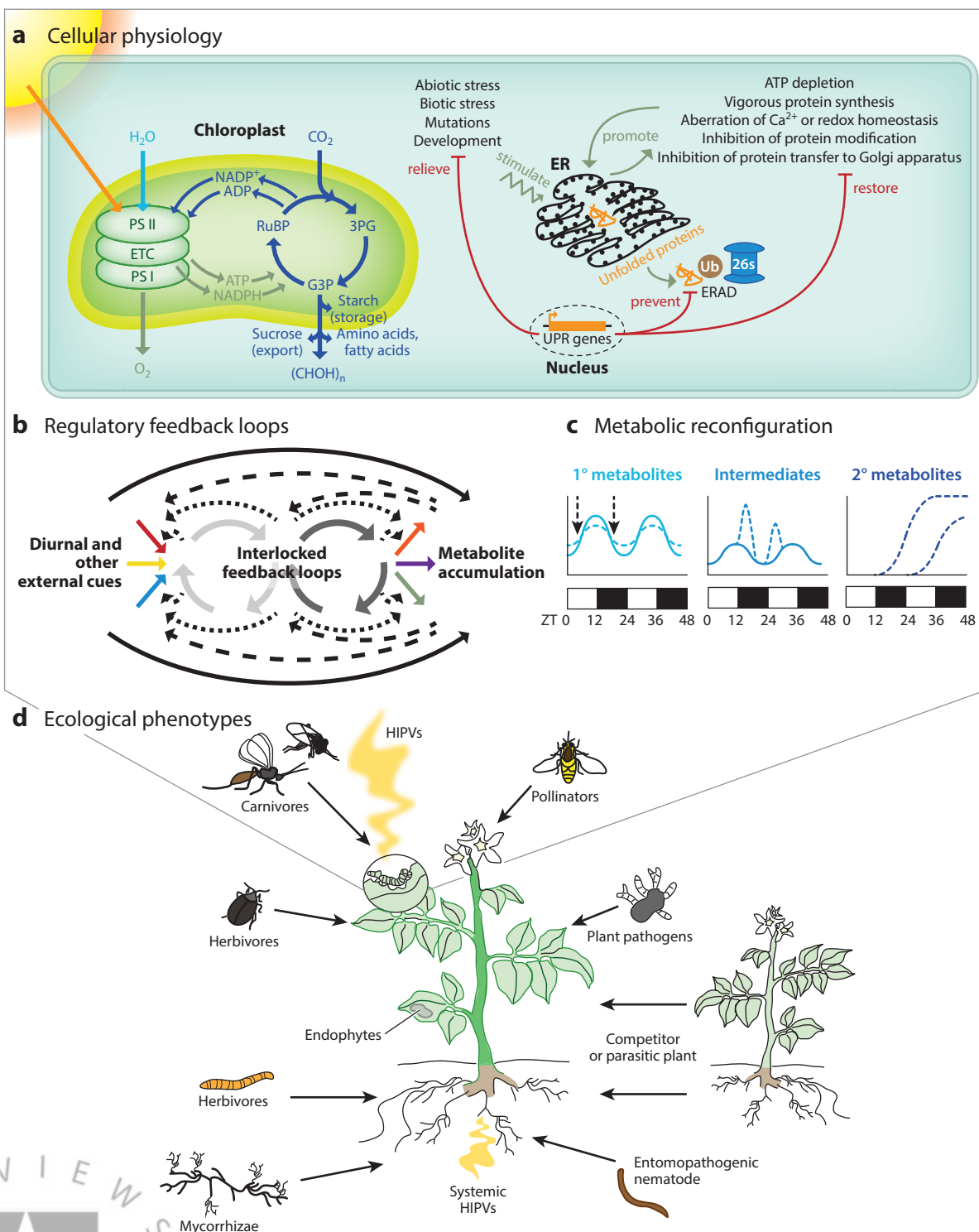
This is important because the role of ecological context in determining phenotypes is poorly understood, most obviously in the case of biological invasions, in which a species out of context dominates its new community. For over two decades, ecologists have sought with little success phenotypic markers of invasiveness (136). Feeney (36) suggested that weediness of invasive plants might be connected to their unapparency in a novel environment, and in 1995, Blossey & Notzold (10) proposed that invasive species evolve increased competitive ability (EICA) when resources are not allocated to defense. Data have not supported this hypothesis, which has misled its field for twenty years (37).

### Where Do Trade-Offs Occur, and How Do They Relate to Function?

The simple trade-off between growth and defense invoked in classical plant defense theories such as OD has been supplanted by an increasingly detailed picture portraying a sophisticated reconfiguration of metabolism in attacked plants, revealed by molecular chemical studies (see,

**SVS:** spatially varying selection





e.g., 8, 9, 12, 54, 94, 110, 122, 125, 147). The individual physiological and ontogenetic trade-offs that contribute to this complex network differ by how much they influence fitness trade-offs at the ecological level. A few master switches regulate many components, and in many cases, molecular interaction studies have already implicated these key players (e.g., 16, 17, 33, 38, 50, 96, 104, 116, 117, 128, 132, 150). However, even when measuring effects of master regulators, the magnitude and even direction of a growth-defense trade-off are likely to depend strongly on environmental variables. Furthermore, traits that attract enemies of herbivores, transport and storage mechanisms that make nutrients inaccessible, and even phenological shifts that confer herbivore avoidance are part of the network regulated by herbivory, and must be considered as components of defense, alongside toxic, repellent, and antinutritive secondary metabolites (8, 36, 64, 69, 112, 114, 115). Defense traits are any traits that defend plant fitness in the face of stress, and the more ecologically realistic the environment, the more robust the demonstration of defensive function (64, 74). As has been discussed elsewhere (e.g., 9, 20), it is often difficult to establish ecologically realistic assays and to measure fitness consequences, but challenges encountered when measuring the right thing cannot be remedied by measuring the wrong thing. We would argue that suitable compromises can be found, so long as these are also clearly defined and their shortcomings acknowledged (e.g., 74, 112).

In fact, we still know astonishingly little about what plants must do in order to grow, defend, reproduce, and keep fit in their native ecologies (**Figure 2**). Although plants are autotrophic organisms, harnessing energy from sunlight and water, fixing carbon from the air, and taking up mineral nutrients from the soil, they also must operate as heterotrophs, metabolizing their fixed carbon and nutrients to form a plethora of other metabolites in a system of precisely coordinated metabolic networks (126). The circadian clock plays a crucial role in regulating these processes, for example, preparing the photosynthetic machinery just before daylight and allowing plants to pace the metabolism of starch during the night (31).

The next section provides an updated overview on the molecular mechanisms of plant defense responses that points out where trade-offs take place, and describes how a multitude of physiological and ontogenetic events are integrated into observed phenotypes, i.e., to incorporate mechanistic, functional, and evolutionary levels of analysis into an integrated understanding without confounding these levels. We continue to take a phytocentric view (8) because this reflects our expertise and because we believe that entomologists benefit from an accurate understanding of plants just as plant scientists benefit from understanding insect herbivores (32, 40). Furthermore, we assume that plants and their herbivores are strategic players in a set of games that play out in ecological time, with consequences for evolutionary trajectories (87, 98). The view we present differs from descriptions of plant defense syndromes (2), which acknowledge the interdependency of defense- and growth-related traits but have not integrated physiological and ecological plasticity.

## Figure 2

An overview of the physiological constraints plants face as photosynthetic organisms when responding to attack from, and manipulating the behavior of heterotrophs: coordination of the light and dark reactions of photosynthesis with (a) subcellular physiology of general and specialized metabolism via the (b) circadian clock, and the resulting (c) chemical and (d) ecological phenotypic parameters that we can measure. Arrows in panel c indicate the timing of an herbivore induction event either at midday (*left arrow*) or midnight (*right arrow*). Abbreviations: ER, endoplasmic reticulum; ERAD, ER-associated degradation; ETC, electron transport chain; HIPVs, herbivore-induced plant volatiles; PSI/II, photosystem I/II; Ub, ubiquitin; UPR, unfolded protein response; ZT, Zeitgeber time (e.g., time since dawn on the first day of the experiment). Images in panel a are modified from References 92 (*left*) and 151 (*right*), image in panel b is modified from Reference 50, and images in panel d are modified from Reference 28. Conceptual graphs in panel c arose from discussions with D. Li (75).

**HAE:**herbivore-associated  
elicitor**FAC:** fattyacid–amino acid  
conjugate**THE LAYERS OF PLANT DEFENSE RESPONSES TO HERBIVORES**

Plant defense responses to herbivores can be understood at any of several layers, from genetic structure through ecological influence (**Table 1**). These layers are separate but not independent: Each emerges from layers below it, and receives feedback from layers above it, but each layer represents a different level of analysis (118). Genetic programming, ontogeny, and environmental factors have all been considered, if not cleanly separated, in plant defense theory. We focus on the molecular physiology of plant responses to insect herbivores, an area in which advances in knowledge have led to an updated experimental and theoretical understanding of plant defense against herbivores. This layer can be divided into (a) perception of herbivores, (b) early signaling responses, (c) hormone-mediated signaling, (d) reconfiguration of metabolism, and (e) changes to phenology. The transcriptional layer of regulation that connects hormonal signaling to phenotypic changes has been reviewed (54) and so we do not repeat it here.

**Perception of Insect Herbivores**

Specificity in response to herbivores begins with plants' perception of damage patterns and herbivore-associated elicitors (HAEs) in amounts as little as femtomolar concentrations (see sidebar, PAMP Env) (reviewed in 12, 33, 60, 93, 147). Plants respond to characteristic spatiotemporal patterns of damage (60, 95) and furthermore can perceive damage to the leaf surface and trichomes when herbivores land or walk (reviewed in 33, 59). Plants seem to count the number of herbivore-associated damage events, and the dynamics of the signaling response to each individual event determine the characteristics of the response to repeated events (reviewed in 41). However, the HAEs found in secretions and fluids from herbivores are essential for many components of plant responses (reviewed in 12, 59, 60, 93, 147).

The first HAEs were discovered in the 1980s and 1990s (reviewed in 12). HAEs may be digested plant molecules such as inceptins, proteolytic products of the plant chloroplastic ATP synthase  $\gamma$ -subunit found in *Spodoptera frugiperda* (fall armyworm) oral secretions (108); or herbivore-derived molecules thought to be essential for metabolism, such as fatty acid–amino acid conjugates (FACs), which are probably required for nitrogen metabolism in Lepidoptera (151). The first fully characterized HAE was the hydroxyl FAC volicitin [*N*-(17-hydroxylinolenoyl)-L-glutamine], identified as the plant-volatile-inducing factor in *Spodoptera exigua* (beet armyworm) oral secretions (3). Data support the existence of a membrane-bound volicitin receptor that is upregulated by herbivory (131), but the protein has not been identified; however, a lectin receptor kinase gene (*LecRK1*) that is rapidly, transcriptionally upregulated in response to the FAC *N*-linolenoyl-glutamate from *Manduca sexta* (tobacco hornworm) was identified in *Nicotiana attenuata* (coyote tobacco) (42). Oral secretions from eight different lepidopteran species furthermore formed ion-permeable channels in artificial membranes, although FACs alone were unable to form stable channels (82).

Plants may further metabolize HAEs as in the case of *N*-linolenoyl-glutamate from *M. sexta*, which is rapidly oxidized by a lipoxygenase upon application to *N. attenuata* leaves (139). This may be one mechanism endowing specificity of plant responses to herbivores. Plant species respond differently to the same HAEs as measured, e.g., by phytohormone production (109), or they may not respond at all to specific HAEs (121). Within plant species, the magnitude and quality of response to the same HAEs vary: In *N. attenuata*, a genotype native to Arizona had a reduced signaling and metabolic response to *M. sexta* oral secretions compared with a genotype from Utah (149), and neighboring plants from single populations in Utah displayed differences of a similar magnitude (63, 113).

**Table 1** Layers of plant defense against herbivores, corresponding levels of analysis, and examples of observed phenomena and testable hypotheses. Levels of analysis are from Sherman (118)

| Layer                                          | Level of analysis      | Phenomena                                                      | Example hypotheses                                                                                                                            |
|------------------------------------------------|------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Ontogenetic changes                            | Ontogeny (proximate)   | Changes in tissue-specific hormonal signaling over life stages | Flowering shuts down herbivore-elicited ethylene signaling.                                                                                   |
|                                                |                        | Tissue- and life-stage-specific accumulation of metabolites    | Herbivore-induced vegetative volatile emission is downregulated when flowers open.                                                            |
|                                                |                        | Stage-specific growth and differentiation patterns             | Sensitivity to herbivore induction as measured by production of hormones, transcripts, or metabolites differs with growth stage.              |
|                                                |                        | Changes in phenology                                           | Senescence processes are induced by herbivory.<br>Flower opening time is changed by herbivory.                                                |
| Physiological changes                          | Physiology (proximate) | Perception of herbivory                                        | Transmembrane receptor proteins detect FACs.                                                                                                  |
|                                                |                        | Early signaling events                                         | The waveform of $[Ca^{2+}]_{cyt}$ fluctuations determines the identity of the protein messenger activated by $Ca^{2+}$ .                      |
|                                                |                        | Phytohormone signaling                                         | JAs other than JA-Ile also bind specific SCF <sup>COI1</sup> -InsP <sub>5</sub> -JAZ complexes to release inhibition of JA-upregulated genes. |
|                                                |                        | Reconfiguration of metabolism                                  | Systemically transported specialized metabolites accumulate in proportion to tissue sink strength.                                            |
|                                                |                        | Changes in growth rate                                         | Different herbivore elicitors result in different magnitudes of reduced growth rates within a plant genotype.                                 |
| Genetic and epigenetic architecture            | Evolution (ultimate)   | Inter- and intrapopulation variation in defense gene alleles   | The frequency of a defensive gene in a plant population is proportional to risk of herbivory by a susceptible herbivore.                      |
|                                                |                        | Epigenetic signatures of herbivory                             | Herbivory in the current generation of plants causes a predictable epigenetic signature in the next generation.                               |
| Ecological interactions and realized phenotype | Function (ultimate)    | Defensive functions of specialized metabolites                 | WT accumulation patterns of specialized metabolites increase plant fitness under herbivory.                                                   |
|                                                |                        | Resistance function of specialized metabolites                 | WT accumulation patterns of specialized metabolites reduce herbivore damage.                                                                  |
|                                                |                        | Fitness costs of producing specialized metabolites             | WT accumulation patterns of specialized metabolites reduce plant fitness in the absence of herbivore attack.                                  |

Abbreviations: FAC, fatty acid–amino acid conjugate; JA, jasmonate; WT, wild type.

## PAMP ENVY

The perception mechanisms for characteristic microbe- or pathogen-associated molecular patterns (MAMPs/PAMPs) and effectors from pathogenic microbes is well understood, having produced the gene-for-gene and guard hypotheses for which there is extensive experimental support (62). This has produced a phenomenon that could be called PAMP envy in which every review of HAEs must begin with a summary of how similar HAEs are to MAMPs, PAMPs, and effectors, yet how relatively little is yet known about HAE perception. This sorry state of affairs is a consequence of where the two fields started: The chemical ecology of plant-insect interactions was developed by zoologists, phytochemists, and plant physiologists, and plant pathology investigations were begun by molecular biologists and geneticists. Plant-insect interaction research generally has a better understanding of the mechanisms, in particular the specific chemistries by which plants and sometimes ecological communities shape interactions, whereas plant pathologists have a more epidemiological understanding: Some genotypes are more resistant than others, but the organism-level mechanism and context dependence of resistance are often not known. An exception is the field of ecological immunology, which attempts to synthesize the cellular, immunological level with the community-based, ecological level of an organism's responses to pathogens (83).

## Early Signaling Response

Upon feeding damage or wounding and application of regurgitant, a membrane potential ( $V_m$ ) depolarization at the feeding site spreads isotropically through the leaf in minutes, as first characterized in detail for *Phaseolus lunatus* (Lima bean) damaged by *Spodoptera littoralis* (Egyptian cotton leafworm) (81). This is rapidly followed by influx of  $\text{Ca}^{2+}$  at the feeding site, which could also be stimulated in *P. lunatus* by wounding and application of volicitin or the detergent sodium dodecyl sulfate (SDS) (81). Within minutes of feeding damage, elevated mitogen-activated protein kinase (MAPK) activity can be detected proximate to the feeding site and in remote locations within the damaged leaf, although the distribution of MAPK activity in *N. attenuata* was constrained by vasculature and vascular flow (148). MAPK activation occurs both up- and downstream of calcium signaling (reviewed in 147), and calcium signaling appears to occur both up- and downstream of reactive oxygen species (ROS) signaling, which is also associated with herbivory but not with mechanical damage; as in pathogen attack, induced ROS production is likely controlled by NADPH oxidases (reviewed in 12, 147).

ROS production, calcium signaling, and MAPK signaling are essential components of responses to a plethora of biotic and abiotic stresses and cues. Thus, the integration of other environmental or stress-related signals likely occurs during the early signaling response to herbivores, but the mechanism of signal integration at this stage is not well understood (reviewed in 12, 147). MAPKs initiate signaling cascades, resulting in activation of transcription factors and phytohormone-mediated signaling (reviewed in 147), and signal integration via phytohormone interactions has been more closely investigated. Interestingly, herbivory-induced MAPK activity patterns correspond to patterns of jasmonate (JA) stress hormone accumulation in *N. attenuata* leaves (127, 148).

### MAPK:

mitogen-activated protein kinase

ROS: reactive oxygen species

JA: jasmonate

## Hormonal Signaling Response

Most herbivore resistance traits are induced by JAs (reviewed in 60, 147), and JAs were shown to negatively regulate tolerance of herbivory in *N. attenuata* (80). JAs are synthesized in plastids and peroxisomes from linolenic acid cleaved from plastidial membranes by glycerolipase (GLA) and peroxidated by lipoxygenase (13). *cis*-(+)-12-Oxo-phytodienoic acid (OPDA), which is synthesized in the plastid, may be the first active hormone in the pathway and is stored, for example, as

arabidopsides for rapid release upon wounding in *Arabidopsis thaliana* (mouse-ear cress) (reviewed in 26). The 18-carbon OPDA is transported to the peroxisome, where it is converted to the 12-carbon (+)-7-iso-jasmonic acid, which may be methylated, hydroxylated, or conjugated to amino acids or sugars to form active and inactive JAs that may be modified further (reviewed in 60, 143, 144).

The conjugate (+)-7-iso-jasmonoyl-L-isoleucine (JA-Ile) is thought to be the primary active form (38); however, JA signaling, but not JA-Ile, is required in *Solanum nigrum* (black nightshade) for resistance against herbivores (138); and JA-Ile is not required for JA-induced volatile emission in *A. thaliana* (137). JA-Ile is perceived by the Skp/Cullin/F-box SCF<sup>COI1</sup> E3 ubiquitin ligase complex (65, 150) interacting with inositol pentakisphosphate (InsP<sub>5</sub>) (117) and one or more jasmonate ZIM-domain (JAZ) proteins (17, 128): Different SCF<sup>COI1</sup>-InsP<sub>5</sub>-JAZ complexes may have altered binding specificities (19, 117). Fifteen JAZ/TIFY proteins in *A. thaliana* have been characterized to date (reviewed in 66), and similar numbers were reported in other plant species (99). The JAZ/TIFY proteins can form complexes with each other and with the JA-regulated transcription factor MYC2 as well as with proteins involved in other hormone signaling pathways (reviewed in 67, 104). In a mechanism similar to auxin and gibberellin signaling, when levels of active JAs are low, JAZ proteins bind to and repress JA-regulated transcription factors; when JA levels are high, the activated SCF<sup>COI1</sup>-InsP<sub>5</sub>-JAZ complex labels JAZ for degradation, activating gene transcription (reviewed in 104). Alternative splicing of JAZ may tune sensitivity by producing variants with reduced affinity for the SCF<sup>COI1</sup> receptor complex (18).

JAs interact extensively with other plant hormones and function in growth and fertility as well as defense (reviewed in 143, 144, 147). The interaction of JAs with other hormones permits integration of multiple stress and developmental signals (reviewed in 33, 104). Ethylene modifies JA-mediated responses, and auxins and the pathogen defense hormone salicylic acid suppress the action of JAs, whereas interactions between JAs and gibberellins (57), cytokinins (106, 107), abscisic acid, and nitric oxide are less well understood (reviewed in 33, 104). Herbivores, similar to pathogens, can manipulate these hormone interactions to their own advantage, often by inducing salicylic acid accumulation to repress JA-mediated responses (reviewed in 104). Some herbivores and pathogens manipulate cytokinins to change the sink strength and nutritional quality of damaged tissues: *Agrobacterium tumefaciens* and galling herbivores use cytokinins to create local sinks in plants, and leafminers form green islands in senescent leaves (reviewed in 33, 43); and more mobile herbivores such as *M. sexta* and *Tupiocoris notatus* (tobacco suckfly) may also manipulate cytokinin signaling (105, 107). Cytokinins from herbivores are often produced by symbiotic microbes (reviewed in 43).

Systemic signaling via the vascular system induces defense production in remote plant parts after attack, similar to the phenomenon of induced systemic resistance after pathogen attack (reviewed in 35, 60, 147). Classical grafting experiments in *Solanum lycopersicum* (tomato) indicated that JA biosynthesis was necessary to produce a long-distance wounding signal, while JA perception was required for its perception in distal leaves, and a JA-derived phloem-mobile signal was proposed (reviewed in 60). Within minutes of wounding, electrical signals spread to systemic tissue via vascular connections and glutamate receptor-like (GLR) ion channels (35, 96, 147). In *A. thaliana*, a lipoxygenase expressed in cells adjacent to xylem is responsible for synthesizing JAs in systemic tissue (16, 35). It is now thought that systemic JA biosynthesis in *A. thaliana* is triggered similarly to JA biosynthesis in distal regions of a wounded leaf: by  $V_m$  depolarization and/or  $Ca^{2+}$  flux transmitted by GLRs and resulting downstream signaling events; however, systemic (between-leaf) signaling is suggested to also involve temporary changes in xylem pressure (35, 81).

The take-home message of the advances in systemic signaling research is that when an herbivore attacks a plant, the entire plant recognizes and responds to this attack at some level, even if



**Specialized metabolism:**

metabolic pathways and products that are not strictly required for growth and production of new cells; also called secondary metabolism

**Direct defense:** plant defense trait that directly affects the ability of an herbivore to assimilate plant tissue, e.g., via a repellent, toxic, or antidigestive mechanism

some responses are highly localized to the attacked tissues. In some plants, priming of systemic defense responses by perception of herbivore-induced volatiles may be a mechanism to avoid vascular constraints on systemic perception of damage (55). The arena of the induced responses is potentially extended to neighboring plants sufficiently close and sensitive to emitted volatiles or that share mycorrhizal connections (7, 55).

### Reconfiguration of Metabolism

Herbivory induces extensive changes to both general and specialized metabolism via a transcriptional reconfiguration specific to each herbivore (reviewed in 54, 114). This reconfiguration can be interpreted as a downregulation of growth and an upregulation of defense in the trade-off framework. Alternatively, it can be understood as the coordinated reconfiguration of metabolism to prevent the allocation of energy and nutrients to herbivore fitness, rather than to plant fitness (**Figure 2, Table 1**). These interpretations are not incompatible, but the second suggests specific hypotheses that are closely connected to known mechanisms of plant physiology and plant-herbivore interactions, and is therefore more useful.

One option for plants to avoid converting their fitness to herbivore fitness is of course to avoid attack, and one way to do this is to “smell bad.” Plant volatiles comprise thousands of structures, and volatile blends are specific to particular stress events and even configurations of herbivores and their parasitization status (28, 100, 119). Volatiles may increase the foraging efficiency of predators and parasitoids of herbivores or prevent herbivores from ovipositing, but herbivores may use the same compounds as cues to locate host plants (reviewed in 28). However, in different ratios, these compounds become more attractive to predators and deterrent to herbivores (4, 5). A volatile blend may comprise the emission from multiple plants, meaning that volatile emission of close neighbors can determine a plant’s attractiveness to herbivores and predators (111). There is evidence that plants can detect neighbors’ volatiles (reviewed in 28, 55), and one yet unexplored function of plants’ sense of smell may be that they adjust their own volatile emission to what the neighbor is producing. Plant volatiles are a double-edged sword.

Once a plant is attacked, the goal of retaining energy and nutrients for plant fitness can be achieved in two ways: by channeling metabolism toward synthesis of secondary metabolites, including direct and indirect defenses, ideally selected for effectiveness against the attacking herbivore; and by enacting tolerance mechanisms to mobilize nutrients away from the attack site and store them in inaccessible forms for later remobilization. The ever-expanding list of known specialized plant metabolites has been reviewed by many others (e.g., 8, 36, 40, 78, 86, 94) and, in addition to more classical examples of plant volatiles, toxins, and digestibility reducers, includes compounds closer to general metabolism such as diterpene glycosides (56, 120) and antinutritive amino acids (1). The discovery that JA signaling dramatically regulates leaf sugar levels (79), in addition to regulating specialized metabolites, underscores how everything in metabolism can be recruited to enhance defense.

Herbivore assimilation rates depend on the composition of nutritive and antinutritive metabolites rather than on the absolute amounts of either: This key insight explained differences in mechanism as well as consequences for digestibility-reducing defenses versus toxic direct defenses proposed by OD and apparency theories (36, 39), and it explains why defense theories such as CNB, RA/GR, and GDB, which at most predict total amounts of secondary metabolites, are not useful. Herbivore consumption rates depend on palatability, which may reflect tissue nutritional quality, but are more likely determined by specific metabolites acting as phagostimulants that may also be important for host location (14, 40, 141), such as the six-carbon green leaf volatiles (46,

47, 90). When it comes to understanding the chemical underpinnings of defense, the devil lies in the chemical details: Composition matters more than total amounts.

The composition of metabolites in tissues can be altered not only by induction of specialized metabolite production and the reallocation of standing pools of antinutritive metabolites, but also by the reallocation of nutritive metabolites through senescence or other processes (discussed in 23, 88, 89, 105). Changes induced by the shoot herbivore *M. sexta* in the expression of the SnRK1 subunit GAL83 increased carbon partitioning to *N. attenuata* roots, reducing loss of energy to herbivory and permitting later remobilization of more carbon from roots for seed production (115). Interestingly, members of the SnRK1 family are involved in regulating plant developmental transitions based on carbon availability and thus may provide a link between growth and defense (132). The ability of plants to reallocate nutritive metabolites will depend on the sink strength of the attacked tissue, which may explain why some (perhaps many) herbivores and pathogens manipulate cytokinin signaling (43, 107).

Particular feeding patterns may also allow herbivores to avoid induced responses in their host plants, akin to the laticifer-cutting behavior employed by herbivores feeding on latex-producing plants (2). By contrast, plant metabolites may manipulate herbivore behavior and susceptibility to attack. For example, ingestion of trypsin protease inhibitors has a small effect on the growth of the specialist herbivore *M. sexta*, but a large effect on its ability to defend against predator attack (112), thus potentially rendering larvae more susceptible to plant indirect defense, although perhaps also rendering them less nutritious prey. Toxic metabolites such as the eponymous alkaloid nicotine in *N. attenuata* may determine the potential predator community, selecting for predators that can avoid ingesting, or that can detoxify, nicotine (72). For the generalist *S. exigua*, consumption of *N. attenuata* tissue is limited by its capacity to detoxify nicotine, which is much less than the capacity of the nicotine-adapted *M. sexta*; this renders *S. exigua* more susceptible to digestibility-reducing protease inhibitors because it cannot compensate by consuming more tissue (124).

The plant circadian clock has been proposed to synchronize the expression of defense traits with herbivore activity (45, 61). However, a key insight from plant defense theory—and one that has proven robust—is that herbivore populations vary with plant chemistry and life history, and any individual plant species is likely to face attack from an herbivore community that varies in time, composition, or both (36, 39). Just as herbivores differ in their ability to cope with plant defense traits, they also differ in the timing of their activity (32, 61). The internal clock tunes sensitivity to diurnally predictable stress factors, but herbivory often does not fit into that category, in contrast to events such as drought stress and pathogen attack, which strongly depend on diurnal changes in abiotic factors such as humidity (61, 116, 142). Thus, the role of the clock in defense against herbivory is likely to be more complex than simple rhythmic control of secondary metabolism. For example, clock regulation of metabolic flux may result in differences in the temporal accumulation of defense metabolites when plants are attacked at different times of day (**Figure 2b**) (75). Stress factors may even disrupt the clock (116, 142).

## Phenology and Timing

OD theory predicts that changes in phenology to reduce apparency could be an aspect of plant defense responses; however, it envisioned these changes over evolutionary and not ecological time (36, 102, 103). More recent studies have shown that growth rate, allocation of resources, flowering time, and even flower phenology are altered in ecological time in response to herbivory (69, 115, 116; and see references in 48, 123). For *N. attenuata*, the specialist herbivore *M. sexta* is also an important pollinator. When attacked by *M. sexta* larvae, the plant changes flowering time

**Specialist herbivore:**  
herbivore that feeds on  
a restricted set of  
plants



from nighttime to morning, thereby switching to likely less effective hummingbird pollination but avoiding further oviposition (69). Leaf removal and elicitation with methyl jasmonate can alter flowering time in competing plants (134); perhaps plants genetically programmed with an earlier flowering time, though likely to be smaller and set less seed, are more likely to successfully distribute seed in seasons with high herbivory pressure, thus stabilizing variation in flowering time in populations.

In *A. thaliana*, natural variation in the sequence of pseudo response regulator (PRR) genes that link external inputs to the internal clock can enhance fitness (91). Variation in the expression of the light-signaling gene *gigantea* (GI) under long-day conditions is widespread and has been proposed to cause conditional variation in the temporal dynamics of diurnally regulated gene expression, which may permit temporal variation in phenotypes while avoiding pleiotropic effects from disrupting the clock (27). Such differences in timing could be mimicked using transgenic plants expressing clock genes under the control of an inducible promoter (105). Sensitivity of plants to herbivores and production of defense metabolites may furthermore fluctuate throughout the day as a result of regulation by the plant circadian clock (reviewed in 116, 146) and also depends strongly on plant developmental stage (11, 29, 49, 71, 135), the progression of which is determined by the clock interacting with environmental cues (24).

## CONCLUSIONS AND FUTURE DIRECTIONS

Plants tailor their defense responses to different herbivores on the basis of HAEs and patterns of damage from, e.g., feeding or oviposition (reviewed in 12, 33, 60, 93). Different plant species respond differently to the same HAEs (109). Within a species, the magnitude and quality of response to an herbivore, as measured by activation of signal transduction components and accumulation of stress hormones, may vary greatly (113, 149), as may the resulting profile of defense metabolites (70; reviewed in 28, 147). What causes this variation? This question can be answered at four different levels of analysis (**Figure 1**).

Ontogenetic changes cause large changes in the hormonal and metabolic response to herbivory. For example, new buds on an oak (*Quercus robur*) will not produce tannins, although the surrounding leaves do (36), and although *M. sexta* feeding on *N. attenuata* produces a burst of ethylene on rosette-stage plants, this burst disappears with the first flower, along with some of the plant's inducible defenses, representing a switch from inducible to constitutive metabolite deployment with the transition to reproductive growth (29)—at least for plants in pots. Mechanistically, different individuals and species have an array of stages in their signaling response to an herbivore that may be tuned toward higher or lower sensitivity, for example, by mutations in *cis*-regulatory DNA elements that control the transcription of regulatory genes, or by mutations in the alternative splicing of JAZ genes and other regulators. At the evolutionary level, balancing selection for two or more alleles of such mutated genes may exist owing to unpredictable or cycling herbivore populations, for example, or plant populations may undergo purifying selection for one allele or another (44). Finally, at the functional level, the effect of any one defense trait on the individual's inclusive fitness will be determined by a changing and interactive Darwinian fitness landscape, meaning that plasticity and honed powers of perception are likely very important, but, as assumed by OD, otherwise anything goes, as long as it is adaptive (98, 102).

Often the evolutionary success of a genotype is determined by its relative advantages compared with those of its neighbors: Can it grow faster, produce more seed, and defend better (87)? However, there are important examples, such as the survival of seeds in the seed bank, in which fitness differences approach categorical distinctions rather than small relative advantages: Is the seed resistant to microbial attack or not? The layered view of plant defense permits separate consideration of

physiological, ontogenetic, ecological, and evolutionary phenomena while recognizing that each layer contains emergent properties from the one(s) below it, and that both diurnal timing and ontogenetic processes are important modifiers of plant responses to the environment. The resulting integrated concept relies on the definition of defense as a set of ecological context-dependent responses that increase plant fitness in the face of herbivore attack. We note that this definition of plant defense is compatible with the concept from ecological immunology of adaptive responses to pathogens (83), and consistent with the attempt to scale up from genes to ecosystems embodied in the community genetics approach (145).

### SUMMARY POINTS

1. Plant defense theory emerged from efforts to understand the distribution of specialized plant metabolites in nature, and the role of these metabolites in structuring ecological communities. However, progress was stymied because the theories frequently posited hypotheses posed at different levels of analysis, which made them impossible to rigorously test.
2. Plant defense phenotypes emerge from the coordination of subcellular and whole-organism physiological processes, and the trade-offs involved are more complex than balancing an account with photosynthetic carbon fixation and nutrient uptake on the one side, and growth and secondary metabolite production on the other.
3. Plant defenses are configured in response to particular attackers via recognition of specific elicitors and patterns of damage, leading to tailored reconfiguration of metabolism that extends to the entire plant for some responses.
4. Specialized plant metabolites are an essential component of plant defense responses, which also include physical barriers, tolerance mechanisms, repellent traits, spatiotemporal escape, and any other traits that increase fitness under herbivore attack.
5. Plant defense responses and their fitness outcomes are shaped by interactions of plants with their ecological communities and their abiotic environment, likely in that order.

### FUTURE ISSUES

1. What are the receptors of HAEs? How can they be identified? Can we learn from plant-pathogen interaction research methods, which have identified multiple receptors of MAMPS and PAMPS and targets of effectors?
2. To what extent is there specificity in the early signaling response to different attackers? How is it shaped by the perception of other stress factors? How do early signaling responses produce tailored phytohormone signaling?
3. How is specificity encoded at the level of phytohormone signaling? What is the significance of the several JAZ/TIFY proteins and their interactions with each other and other proteins? What role do JAs other than JA-Ile play? How are multiple hormone signaling pathways integrated? How are they manipulated by herbivores?



4. What are the roles of general and specialized metabolism in determining nutritional quality and palatability for more or less specialized insect herbivores? How do direct and indirect defense strategies interact? How is this metabolic reconfiguration constrained by ontogeny? What are the cellular constraints on launching induced defenses?
5. How do the phenotypes of neighbors alter a plant's own realized phenotype and fitness landscape? What are the consequences for biodiversity and ecosystem function?
6. How can we use this knowledge to address pressing practical concerns such as biological invasions and agricultural sustainability?
7. Can we avoid hypothesis conflation, unfalsifiability, and "talking past each other" in the future by applying Sherman's levels of analysis and conceptual frameworks based on these levels, such as the concept of plant defense as a series of layers?

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